

VILLACH STADT, Austria

WMO: 112130

Lat: 46.618N

Lon: 13.874E

Elev: 495

StdP: 95.52

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.3	-9.3	-13.9	1.2	-7.6	-11.9	1.4	-5.5	5.9	4.4	5.2	4.8	0.7	230	0.460

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	31.0	20.5	29.2	19.9	27.5	19.1	21.5	29.3	20.6	27.7	19.7	26.1	2.6	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.8	14.5	24.4	18.1	13.8	23.4	17.4	13.2	22.5	64.6	29.3	61.3	27.8	58.4	26.3	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
5.5	4.6	3.9	DB	-13.6	33.9	3.2	2.0	-15.9	35.3	-17.7	36.5	-19.5	37.6	-21.8	39.1	
			WB	-13.9	23.0	3.1	0.8	-16.1	23.6	-17.9	24.1	-19.6	24.5	-21.9	25.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	-1.4	1.2	5.7	10.6	15.1	18.9	20.7	20.0	15.2	10.6	4.7	-0.6
	DBStd	8.49	3.73	3.56	3.92	3.75	3.44	3.59	3.16	3.16	3.25	3.77	3.82	3.43
	HDD10.0	1309	353	246	140	37	3	0	0	0	2	39	161	328
	HDD18.3	3233	612	479	392	231	110	37	14	18	103	242	408	586
	CDD10.0	1348	0	0	6	56	162	266	331	311	157	57	3	0
	CDD18.3	228	0	0	0	0	11	53	86	71	8	0	0	0
	CDH23.3	1878	0	0	0	9	111	454	714	538	51	1	0	0
	CDH26.7	546	0	0	0	1	17	130	234	161	3	0	0	0
(14) Wind	WSAvg	1.5	1.2	1.3	1.6	1.7	1.8	1.7	1.8	1.6	1.4	1.3	1.3	1.1
(15) Precipitation	PrecAvg	1285	51	52	70	89	104	132	150	147	136	140	139	76
	PrecMax	1691	221	232	169	149	199	227	295	221	236	342	458	234
	PrecMin	994	2	0	3	20	37	49	45	56	56	0	15	10
	PrecStd	185	44	52	43	38	36	44	52	48	50	77	96	52
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.3	13.7	19.9	24.6	29.0	32.2	33.7	33.2	26.9	22.8	15.9	9.1
		MCWB	6.3	7.7	10.9	14.5	18.8	20.8	22.0	21.3	19.0	15.9	11.0	6.4
	2%	DB	7.1	10.6	17.2	21.9	26.2	30.0	31.1	30.4	24.8	20.1	13.3	6.9
		MCWB	3.9	5.5	9.7	13.0	17.0	20.2	20.7	20.4	17.8	14.6	10.1	5.0
	5%	DB	5.1	8.6	14.8	20.0	24.2	28.0	29.2	28.3	22.8	18.2	11.4	5.0
		MCWB	2.8	4.4	8.5	12.0	15.8	19.1	20.0	19.8	16.7	13.6	9.4	3.4
	10%	DB	3.5	6.7	12.6	17.8	22.2	26.1	27.4	26.4	20.9	16.2	9.9	3.7
		MCWB	1.9	3.3	7.3	11.1	14.8	18.1	19.2	19.0	15.6	12.6	8.5	2.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.3	8.1	11.5	15.0	19.3	22.1	22.7	22.3	19.6	16.4	12.0	7.1
		MCDB	9.5	13.0	18.5	23.5	27.1	30.2	31.9	31.0	25.6	21.4	13.7	8.3
	2%	WB	4.6	6.1	10.0	13.7	17.7	20.9	21.6	21.1	18.4	15.2	10.8	5.3
		MCDB	6.5	9.7	16.5	20.5	24.7	28.4	29.5	28.7	23.6	19.1	12.5	6.5
	5%	WB	3.2	4.8	8.9	12.6	16.5	19.8	20.6	20.3	17.4	14.2	9.7	3.8
		MCDB	4.7	8.0	14.3	18.5	22.8	26.5	27.7	27.1	21.6	17.3	11.3	4.7
	10%	WB	2.1	3.6	7.8	11.6	15.4	18.8	19.7	19.4	16.4	13.2	8.6	2.7
		MCDB	3.2	6.3	11.9	16.7	21.0	24.6	26.0	25.3	19.9	15.9	9.9	3.5
(35) Mean Daily Temperature Range	5% DB	MDBR	6.6	8.7	10.7	10.5	10.8	11.0	11.4	10.6	9.6	8.6	5.9	5.6
		MCDBR	9.0	12.3	14.9	14.5	14.3	14.2	14.0	13.8	11.9	11.2	8.3	7.6
	5% WB	MCWBR	6.0	7.4	8.0	6.8	5.9	5.6	5.2	5.3	5.5	6.0	5.3	5.4
		MCDWR	7.7	11.3	13.9	12.8	13.1	13.3	13.0	12.7	10.8	10.1	7.0	6.2
(40) Clear-Sky Solar Irradiance	taub	taub	0.298	0.329	0.381	0.432	0.442	0.451	0.447	0.440	0.408	0.381	0.340	0.300
		taud	2.379	2.300	2.220	2.136	2.165	2.192	2.213	2.247	2.307	2.348	2.410	2.440
	Ebn at Noon	Ebn at Noon	794	833	835	822	827	820	818	807	799	763	733	752
		Edh at Noon	72	95	120	143	144	141	137	127	110	91	71	61
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.40	2.24	3.37	4.23	5.01	5.61	5.69	4.99	3.67	2.44	1.39	1.17
	RadStd	RadStd	0.17	0.31	0.42	0.53	0.57	0.43	0.42	0.51	0.45	0.28	0.24	0.16

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (89 neighbors)	+0.48	N/A	N/A	+0.72	+0.42	+0.33	N/A	-124	+88	+55

Nomenclature: See separate page